

Polar Unit Test (Practice E)

Calculator

You have the entire period.

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test. _____

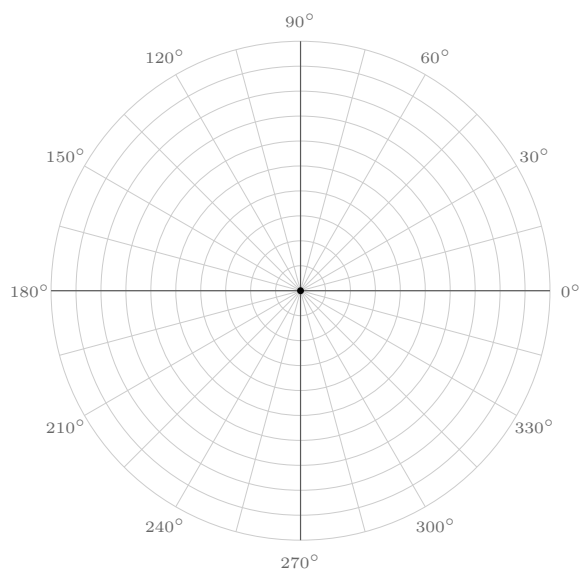
SHOW ALL WORK FOR FULL CREDIT.

Signature _____

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. Show your set up (what you are putting into the calculator). (3 points each)

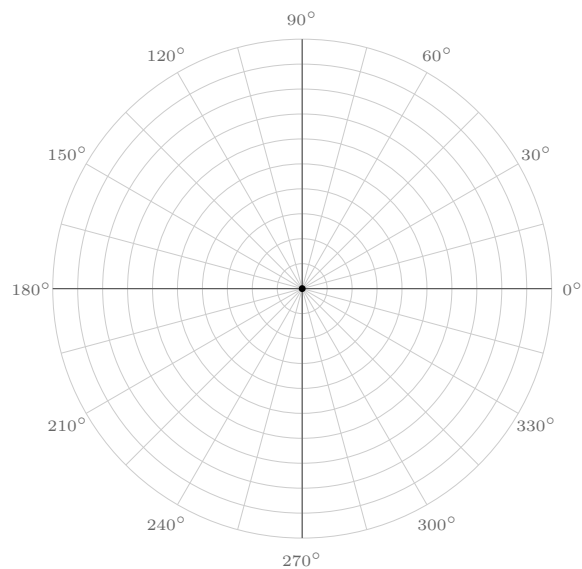
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 2 + 4 \sin \theta$



Type: _____

2. $r = 6 \sin 4\theta$



Type: _____

Polar Unit Test (Practice E)

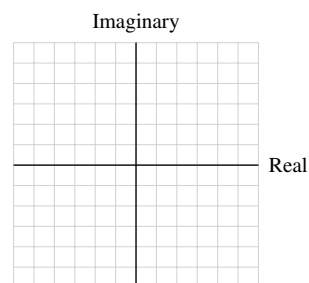
Non-Calculator

SHOW ALL WORK FOR FULL CREDIT.

Simplify. (3 points) 3. $\frac{\sqrt{15}}{\sqrt{-5} \cdot \sqrt{-6}}$	Simplify. (2 points) 4. $\frac{-6 + \sqrt{-27}}{3}$
Perform the indicated operation. Write the answer in rectangular form. (2 points) 5. i^{-13}	Perform the indicated operation. Write the answer in rectangular form. (2 points) 6. $(3 - 8i) - (-5 + 2i) - (4 - 6i)$
Perform the indicated operation. Write the answer in rectangular form. (3 points) 7. $\frac{3 + 4i}{2 - i}$	Perform the indicated operation. Write the answer in rectangular form. (3 points) 8. $i(4 - i)(2 + 3i)$
Solve over the set of complex numbers. (3 points) 9. $x^2 + 2x + 10 = 0$	Convert to trig form. (3 points) 10. $-4 - 4i$

Find the sum. Write it in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (3 points)

11. $(5 - 2i) + (-2 + 6i)$



Using $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$ is the following in the Julia set? Show ALL work. (3 points)

12. $z = 1 - i$

YES / NO

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{10(\cos 50^\circ + i \sin 50^\circ)}{5(\cos 320^\circ + i \sin 320^\circ)}$

14. $[\sqrt{3} \operatorname{cis} \frac{11\pi}{6}] [2 \operatorname{cis} \frac{5\pi}{6}]$

Using $z, z^2 + z, (z^2 + z)^2 + z, \dots$ is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = 2i$

YES / NO

Find each power. Write your answer in rectangular form. (3 pts)

16. $(-1 + i\sqrt{3})^5$

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^3 + 8 = 0$

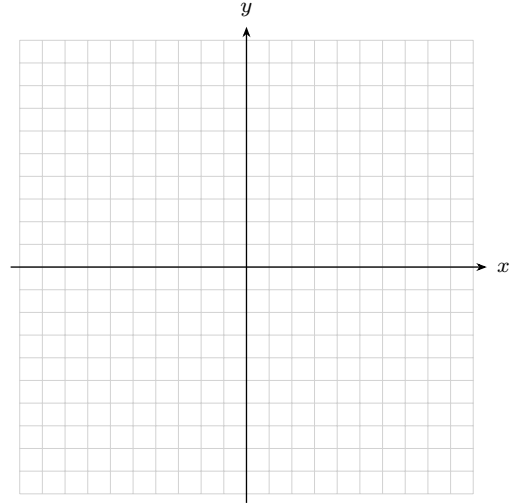
Give two other pairs of polar coordinates for the point: one with $r > 0$ and one with $r < 0$, with $-360^\circ \leq \theta < 360^\circ$. (3 points) 18. $(5, 120^\circ)$	Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points) 19. $(-4, 225^\circ)$
Convert the rectangular equation to a polar equation. Solve for r. (3 points) 20. $x^2 + y^2 = 8x$	Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points) 21. $(3\sqrt{2}, 45^\circ)$ and $(6, 270^\circ)$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = \frac{2}{1 - \sin \theta}$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = 4 \cos \theta + 4 \sin \theta$



Find all points of intersection for $0 \leq \theta < 2\pi$. Include the pole if the curves share it. (4 points)

24. $r = 2 + 2 \cos \theta$, $r = 2 - 2 \cos \theta$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r^2 = 9 \sin 2\theta$

Type: _____

ANSWER KEY – Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Pace check: about 50 minutes for a strong pass. Calculator part about 5, #3–12 about 15, #13–17 about 13, #18–25 about 15. That leaves about 30 minutes of the period to check your work.

Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

1. $r = 2 + 4 \sin \theta$

WORK

Polar mode: $r1(\theta) = 2 + 4 \sin \theta$

$90^\circ: r = 6; 0^\circ, 180^\circ: r = 2; 270^\circ: r = -2$

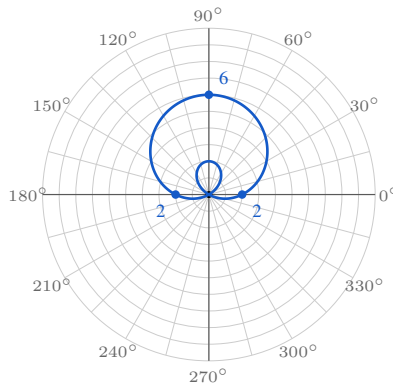
$$\frac{a}{b} = \frac{2}{4} \neq 1$$

limaçon

WHY

Degree mode needs θ from 0 to 360 (radians: 0 to 2π).

Negative r at 270° plots up at $(0, 2)$: inner loop.



Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

2. $r = 6 \sin 4\theta$

WORK

Polar mode: $r1(\theta) = 6 \sin(4\theta)$

$n = 4$ even: 8 petals, each 6 long

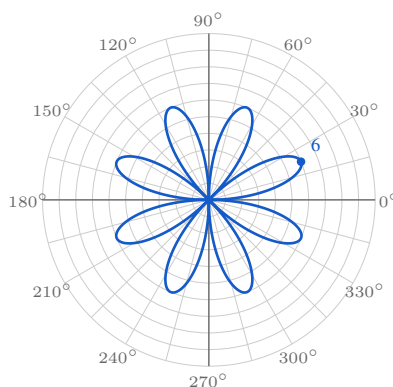
first tip at 22.5° , then every 45°

rose curve

WHY

$2n$.

$\sin 4\theta = 1$ at $4\theta = 90^\circ$.



ANSWER KEY – Non-Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Simplify. (3 points)

3. $\frac{\sqrt{15}}{\sqrt{-5} \cdot \sqrt{-6}}$

WORK

$$\frac{\sqrt{15}}{i^2 \sqrt{30}} = -\frac{1}{\sqrt{2}}$$

$$\boxed{-\frac{\sqrt{2}}{2}}$$

WHY

Pull out i first.

Rationalize.

Simplify. (2 points)

4. $\frac{-6 + \sqrt{-27}}{3}$

WORK

$$\sqrt{-27} = 3i\sqrt{3}$$

$$\frac{-6 + 3i\sqrt{3}}{3} = \boxed{-2 + i\sqrt{3}}$$

WHY

Perform the indicated operation. Write the answer in rectangular form. (2 points)

5. i^{-13}

WORK

$$-13 + 16 = 3$$

$$i^3 = \boxed{0 - i}$$

Trap: i^{-13} is not $-i^{13}$. Negative power means reciprocal.

WHY

Add a multiple of 4 until the power is positive.

Check: $\frac{1}{i^{13}} = \frac{1}{i} = \frac{1}{i} \cdot \frac{-i}{-i} = -i$.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

6. $(3 - 8i) - (-5 + 2i) - (4 - 6i)$

WORK

$$3 - 8i + 5 - 2i - 4 + 6i$$

$$\boxed{4 - 4i}$$

WHY

Perform the indicated operation. Write the answer in rectangular form. (3 points)

7. $\frac{3+4i}{2-i}$

WORK
 $\frac{2+i}{2+i}$

top: $6+3i+8i+4i^2 = 2+11i$

bottom: $4+1 = 5$

$$\frac{2}{5} + \frac{11}{5}i$$

Trap: The bottom is $2^2 + 1^2 = 5$, not $4 - 1 = 3$.

WHY

Conjugate of the bottom.

$$a^2 + b^2.$$

Split into two fractions.

Perform the indicated operation. Write the answer in rectangular form. (3 points)

8. $i(4-i)(2+3i)$

WORK

$$(4-i)(2+3i) = 8+12i-2i-3i^2 = 11+10i$$

$$i(11+10i) = \boxed{-10+11i}$$

WHY

Solve over the set of complex numbers. (3 points)

9. $x^2 + 2x + 10 = 0$

WORK

$$x = \frac{-2 \pm \sqrt{4-40}}{2} = \frac{-2 \pm 6i}{2}$$

$$\boxed{x = -1 \pm 3i}$$

WHY

Convert to trig form. (3 points)

10. $-4-4i$

WORK

$$r = 4\sqrt{2}, \text{ ref } 45^\circ, \text{ QIII: } 225^\circ$$

$$\boxed{4\sqrt{2}(\cos 225^\circ + i \sin 225^\circ)}$$

WHY

Find the sum. Write it in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (3 points)

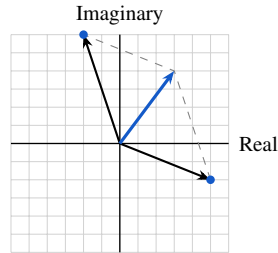
11. $(5 - 2i) + (-2 + 6i)$

WORK

$3 + 4i$

$|3 + 4i| = \sqrt{9 + 16} = 5$

Graph: $(5, -2)$, $(-2, 6)$, resultant $(3, 4)$



WHY

Add real parts, add imaginary parts.

$\sqrt{a^2 + b^2}$.

Resultant is the parallelogram's diagonal.

Using $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1$, ... is the following in the Julia set? Show ALL work. (3 points)

12. $z = 1 - i$

YES / NO

WORK

$(1 - i)^2 - 1 = -2i - 1 = -1 - 2i$

$|-1 - 2i| = \sqrt{5} \approx 2.24 > 2$

NO

Trap: $(1 - i)^2$ is $-2i$, not $1 + 1$.

WHY

$(1 - i)^2 = 1 - 2i + i^2 = -2i$.

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

13. $\frac{10(\cos 50^\circ + i \sin 50^\circ)}{5(\cos 320^\circ + i \sin 320^\circ)}$

WORK

$2 \operatorname{cis}(-270^\circ) = 2 \operatorname{cis} 90^\circ$

$0 + 2i$

WHY

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

14. $\left[\sqrt{3} \operatorname{cis} \frac{11\pi}{6}\right] \left[2 \operatorname{cis} \frac{5\pi}{6}\right]$

WORK

$2\sqrt{3} \operatorname{cis} \frac{16\pi}{6} = 2\sqrt{3} \operatorname{cis} \frac{8\pi}{3}$

$\frac{8\pi}{3} - 2\pi = \frac{2\pi}{3}$

$2\sqrt{3} \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -\sqrt{3} + 3i$

WHY

Using $z, z^2 + z, (z^2 + z)^2 + z, \dots$ is the following number in the Mandelbrot set? Show ALL work. (3 points)

15. $z = 2i$

YES / NO

WORK

$$|2i| = 2$$

$$(2i)^2 + 2i = -4 + 2i, \sqrt{20} > 2$$

WHY

Not over 2 yet.

Find each power. Write your answer in rectangular form. (3 pts)

16. $(-1 + i\sqrt{3})^5$

WORK

$$r = 2, \text{ ref } 60^\circ, \text{ QII: } 120^\circ$$

$$32 \text{ cis } 600^\circ = 32 \text{ cis } 240^\circ$$

$$32 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \boxed{-16 - 16\sqrt{3}i}$$

WHY

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17. $x^3 + 8 = 0$

WORK

$$x^3 = -8 = 8 \text{ cis } 180^\circ$$

$$2, 60^\circ + 120^\circ k$$

$$\boxed{2 \text{ cis } 60^\circ, 2 \text{ cis } 180^\circ, 2 \text{ cis } 300^\circ}$$

WHY

Trap: $x = -2$ is only one of the three roots.

Give two other pairs of polar coordinates for the point: one with $r > 0$ and one with $r < 0$, with $-360^\circ \leq \theta < 360^\circ$. (3 points)

18. $(5, 120^\circ)$

WORK

$$r > 0: 120^\circ - 360^\circ = -240^\circ: \boxed{(5, -240^\circ)}$$

$$r < 0: 120^\circ + 180^\circ = 300^\circ: \boxed{(-5, 300^\circ)} \text{ or } (-5, -60^\circ)$$

WHY

Same r : add or subtract 360° .

Flip r : add or subtract 180° .

Trap: Changing the sign of r without moving the angle by 180° gives the opposite point.

Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points)

19. $(-4, 225^\circ)$

WORK

$$x = -4 \left(-\frac{\sqrt{2}}{2} \right) = 2\sqrt{2}, y = 2\sqrt{2}$$

$$\boxed{(2\sqrt{2}, 2\sqrt{2})}$$

WHY

Opposite of QIII: QI.

Convert the rectangular equation to a polar equation. Solve for r . (3 points)

20. $x^2 + y^2 = 8x$

WORK

$$r^2 = 8r \cos \theta$$

$$r = 8 \cos \theta$$

WHY

Substitute.

Divide by r .

Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points)

21. $(3\sqrt{2}, 45^\circ)$ and $(6, 270^\circ)$

WORK

$(3, 3)$ and $(0, -6)$

$$m = \frac{3 - (-6)}{3 - 0} = 3, \quad y = 3x - 6$$

$$3x - y = 6$$

$$r = \frac{6}{3 \cos \theta - \sin \theta}$$

WHY

$$3\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 3.$$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

22. $r = \frac{2}{1 - \sin \theta}$

WORK

$$r - r \sin \theta = 2, \quad r = y + 2$$

$$x^2 + y^2 = y^2 + 4y + 4$$

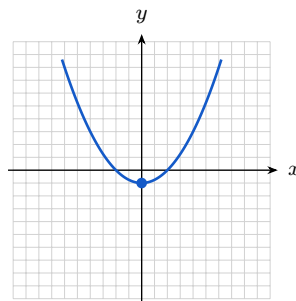
$$x^2 = 4(y + 1)$$

WHY

Clear fraction; $r \sin \theta = y$.

Square both sides.

Parabola, vertex $(0, -1)$, opens up.



Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23. $r = 4 \cos \theta + 4 \sin \theta$

WORK

$$x^2 + y^2 = 4x + 4y$$

$$(x^2 - 4x + 4) + (y^2 - 4y + 4) = 8$$

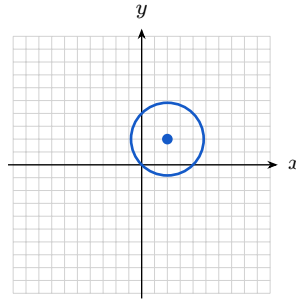
$$(x - 2)^2 + (y - 2)^2 = 8$$

WHY

Multiply by r .

Complete the square twice.

Center $(2, 2)$, radius $2\sqrt{2}$.



Find all points of intersection for $0 \leq \theta < 2\pi$. Include the pole if the curves share it. (4 points)

24. $r = 2 + 2 \cos \theta$, $r = 2 - 2 \cos \theta$

WORK

$$\cos \theta = 0: \theta = \frac{\pi}{2}, \frac{3\pi}{2}, r = 2$$

First curve: $r = 0$ at π ; second: $r = 0$ at 0

$$\left(2, \frac{\pi}{2}\right), \left(2, \frac{3\pi}{2}\right), \text{ the pole}$$

WHY

Set $r = r$.

Both reach the pole, at different angles.

Trap: Setting $r = r$ never finds the pole. Check each curve for $r = 0$.

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25. $r^2 = 9 \sin 2\theta$

WORK

lemniscate

WHY

$$r^2 = a^2 \sin 2\theta.$$